

AI Assisted Coding Assignment- 18.3

2303A51886 : BT: 30 SAI SPURTHI. B 31.03.2026

Lab 18 – API Integration: Connecting to External Services
with Error Handling

Lab Objectives:

- Learn how to integrate Python programs with external REST APIs.
- Understand API request/response handling using requests or similar libraries.
- Implement proper error handling for failed API calls (timeouts, invalid responses, rate limits).
- Practice extracting and displaying meaningful information from API responses.

Task 1 – Public Transport Route Fare API

Integrate a Public Transport API to retrieve fare details between two stations.

Instructions:

- Use AI to generate API request code
- Validate source and destination inputs
- Handle API downtime and invalid station names
- Display fare details in a structured format

Expected Output:

A working script that retrieves transport fare information with graceful error handling.

```
Lab-18.3-ASS.py > get_fare
1 #Task1
2 def get_fare(source, destination):
3     if not source or not destination:
4         print("Invalid input")
5         return
6     # Dummy fare logic
7     fare_data = {
8         ("Chennai", "Bangalore"): 120,
9         ("Chennai", "Hyderabad"): 200,
10        ("Delhi", "Mumbai"): 300
11    }
12    try:
13        fare = fare_data.get((source, destination))
14        if fare:
15            print("\nFare Details:")
16            print(f"From: {source}")
17            print(f"To: {destination}")
18            print(f"Fare: ₹{fare}")
19        else:
20            print("Invalid station names")
21    except Exception:
22        print("Something went wrong")
23
24 # Test
25 get_fare("Chennai", "Bangalore")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + - [] [X] ...

```
i bellamkonda\OneDrive\Desktop\AI CODING\Lab-18.3-ASS.py"
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurthi bellamkonda/OneDrive/Desktop/AI CODING/Lab-18.3-ASS.py"
```

Fare Details:
From: Chennai
To: Bangalore
Fare: ₹120

```
Lab-18.3-ASS.py •
Lab-18.3-ASS.py > convert_currency
28 import requests
29 def convert_currency(amount):
30     if amount <= 0:
31         print("Invalid amount")
32         return
33     url = "https://api.exchangerate-api.com/v4/latest/INR"
34     try:
35         res = requests.get(url, timeout=5)
36         if res.status_code != 200:
37             print("API error")
38             return
39         data = res.json()
40         rates = data["rates"]
41         print("\nConversion Table:")
42         print(f"{'Currency':<10} {'Amount'}")
43         print("-----")
44         print(f"USD          {amount * rates['USD']:.2f}")
45         print(f"EUR          {amount * rates['EUR']:.2f}")
46         print(f"GBP          {amount * rates['GBP']:.2f}")
47     except requests.exceptions.Timeout:
48         print("Timeout error")
49     except Exception as e:
50         print("Error:", e)
51 # Test
52 convert_currency(1000)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurthi bellamkonda/OneDrive/Desktop/AI CODING/Lab-18.3-ASS.py"

Conversion Table:
Currency Amount
-----
USD      10.60
EUR       9.22
GBP       8.01
```



```
Lab-18.3-ASS.py •
Lab-18.3-ASS.py > get_news
--
82 #Task-4
83 import requests
84 import time
85 API_KEY = "YOUR_API_KEY"
86 def get_news(category):
87     url = f"https://newsapi.org/v2/top-headlines?category={category}&apiKey={API_KEY}"
88     for attempt in range(2): # retry mechanism
89         try:
90             res = requests.get(url, timeout=5)
91             if res.status_code != 200:
92                 print("Error fetching news")
93                 return
94             data = res.json()
95             articles = data.get("articles", [])[:5]
96             print("\nTop Headlines:")
97             for i, article in enumerate(articles, 1):
98                 print(f"{i}. {article['title']}")
99             return
100         except requests.exceptions.RequestException:
101             print("Retrying...")
102             time.sleep(2)
103     print("Failed after retry")
104 # Test
105 get_news("technology")
106
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\spurti bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurti bellamkonda/OneDrive/Desktop/AI CODING/Lab-18.3-ASS.py"
Forks: 61255
Open Issues: 3
Error fetching news
○ Forks: 61255
Open Issues: 3
Error fetching news
Error fetching news
```

Ln 103, Col 32 Spaces: 4 UTF-8 CRLF {} Python Python 3.8 (64-bit) Go

Task 5 -COVID-19 Statistics API Integration

Task:

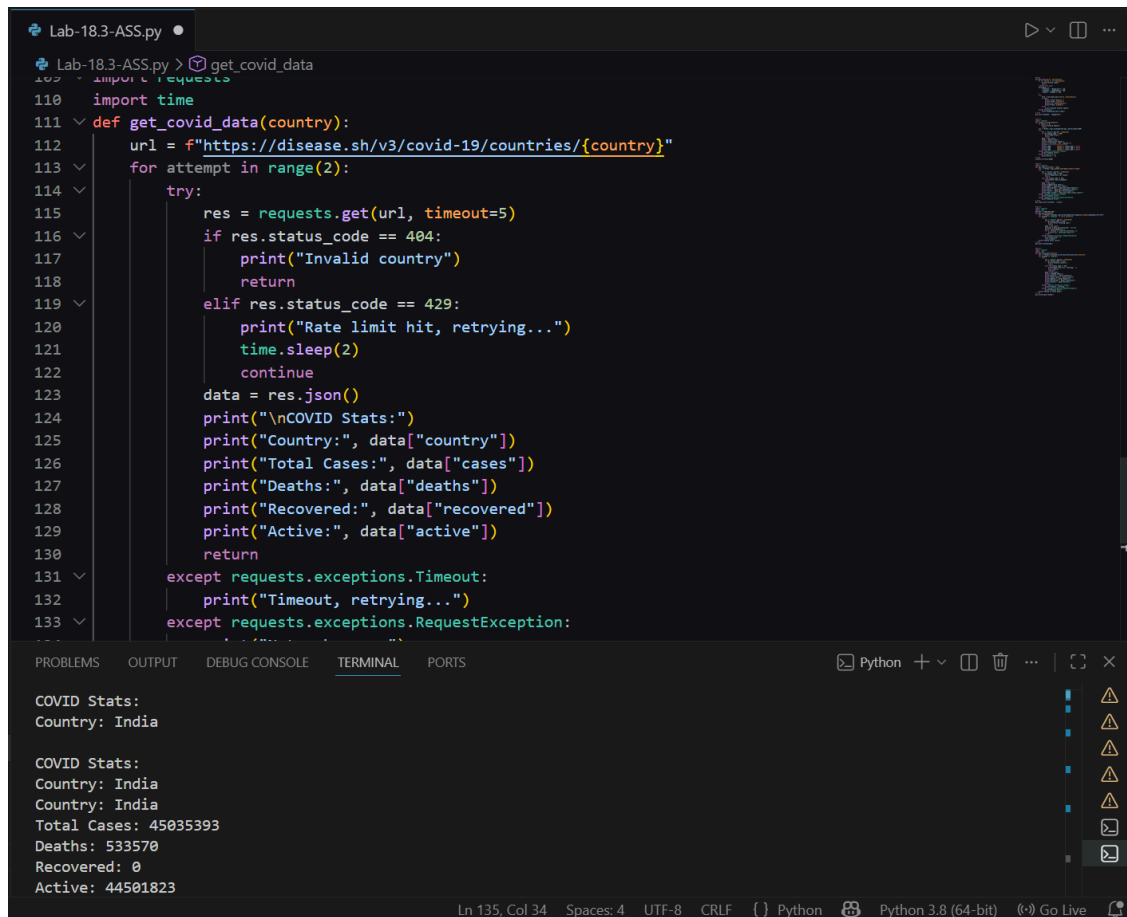
Connect to a COVID-19 Statistics API to fetch country-wise COVID data.

Instructions:

- Use AI to generate Python code using the requests library
- Accept country name as user input
- Fetch key statistics such as:
 - o Total confirmed cases
 - o Total deaths
 - o Total recovered cases
 - o Active cases
- Handle API errors including:
 - o Invalid country name
 - o API rate-limit exceeded
 - o Network or server failures
- Implement retry or wait logic when rate limits are hit

Expected Output:

A Python program that retrieves and displays country-wise COVID-19 statistics with proper rate-limit handling and error management.



The screenshot shows a VS Code editor window with a file named 'Lab-18.3-ASS.py'. The code defines a function `get_covid_data(country)` that uses the `requests` library to fetch COVID-19 data from a specific URL. It includes error handling for `404` (Invalid country), `429` (Rate limit hit), and `Timeout` exceptions. The function prints the country name and its COVID-19 statistics: Total Cases, Deaths, Recovered, and Active cases. The terminal output shows the function being called for 'India', resulting in the following statistics:

```
COVID Stats:
Country: India
Country: India
Country: India
Total Cases: 45035393
Deaths: 533570
Recovered: 0
Active: 44501823
```